



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Engineering

Academic Year 2022 – 2023 (Odd Semester)

Degree, Semester & Branch: I Semester B.E. CSE 'A'

Course Code & Title: GE3151 Problem Solving and Python Programming

Name of the Faculty member (s): Mrs.B.Vijayalakshmi, AP (SG)/CSE

Innovative Practice Description

- Unit / Topic: Unit I / Illustrative Problem: Towers of Hanoi**

- Course Outcome: CO1**

- Topic Learning Outcome: TLO3**

- Activity Chosen: Role play**

- Justification:**

The role play helps the students to play and understand the basics of theories handled. The solving of Towers of Hanoi is an interesting game but the recursion concept behind this problem is difficult to understand. In order to provide the clear understanding of recursion the roleplay activity was conducted.

- Time Allotted for the Activity: 10 Minutes**

- Details of the Implementation:**

- Role play is used to show a live demonstration of how a particular task will be handled. It gives a clear understanding to the students while they are involving in that activity.
- The faculty member explained the concept of solving the Towers of Hanoi problem.
- 3 students are asked to play the role of source tower, destination tower and auxiliary tower. The game rules are explained to them. They have to follow the rules to solve the problem.
- Student 1 (Mr.Enbasankar) act as a source tower has 3 objects as shown in Figure 1. Based on the rule the object should be moved to destination tower (student 3- Mr.Asrar Siddiq) with the help of auxiliary tower (student 2- Mr.Thiruramkumar) in the same order.
- Figure 2 shows the movement of disk from 1 source to destination and the disk 2 to auxillary tower and the third disk in source tower.
- Figure 3 shows the final stage where all the disks are moved to destination tower.
- From this the students clearly understood the concept of recursion clearly.
- Figure 1 shows the students involvement in the activity.

- CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO5	PO8	PO9	PO10	PO12	PSO1	PSO2	PSO3
CO5	3	2	2	2	1	1	1	1	3	2	3

(1 – Low 2 – Moderate 3 – High)

- PO / PSO mapped:**

	PO1	PO2	PO3	PO9	PO10	PSO1
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Innovative practice	3	2	2	1	1	2
Justification for correlation	The basic knowledge about recursion is needed	The students analyze the problem for solution finding solve.	The students find solution to solve the problem.	The students coordinate and demonstrate the activity as a team.	The students communicate how the problem solved.	The recursion concept is used in developing the software

- **Images / Screenshot of the practice:**



Figure 1: 3 Disks in Source tower (Mr.Enba Sankar)



Figure 2: Disk 1 in destination tower (Mr.Asrar Siddiq), Disk 2 in auxiliary tower(MR.Thiruramkumar) & Disk 3 in Source tower (Mr.Enba Sankar)



Figure 1: 3 Disks in Destination tower (Mr.Asrar Siddiq)

- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

- They feel it is very interesting and told they understand how the to solve the towers of Hanoi.
 - The students are energetically involved and enjoyed the activity.

- ❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- It helps the students to understand the recursive procedure of solving the towers of Hanoi problem.
 - This activity makes the class more interactive.

- ❖ **Challenges faced in implementation:**

- All the students cannot be involved in the activity.

References:

- <https://www.ritrjpm.ac.in/images/computer-science/CS8493-OS-Roleplay.pdf>
- https://www.ritrjpm.ac.in/images/computer-science/38_CS8391_Roleplay.pdf

Signature of Faculty Member

HOD